

ELEC73149 – Artificial Intelligence

University of Staffordshire | Department of Engineering

Problem Solving using Advanced Fuzzy Logic and Machine Learning

Early Prediction of Septic Shock Risk: A Weighted Hybrid Fuzzy–Machine Learning System

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ABSTRACT

Septic shock is a rapidly progressive and life-threatening condition in which subtle physiological deterioration precedes haemodynamic collapse. Conventional ICU alarm systems rely on rigid thresholds and often fail to detect early warning patterns. This study develops a weighted hybrid intelligent system that integrates a Mamdani fuzzy inference system (FIS) with supervised machine-learning (ML) models to estimate septic-shock risk from continuous physiological data. Seven clinically grounded vital-sign features were selected from a 144,000-record dataset based on Sepsis-3 criteria and data-quality considerations. Because the dataset lacked explicit labels, a physiologically validated proxy target was engineered, resulting in an extreme imbalance (0.042% positive cases). Three supervised models—Logistic Regression, Random Forest and LightGBM—were evaluated, with LightGBM achieving the strongest credible performance.

The hybrid architecture fuses the normalised fuzzy-risk score with the LightGBM probability using a weighted scheme,

$$H = 0.4F_{norm} + 0.6P_{ML},$$

where F_{norm} is the normalised FIS risk score and P_{ML} is the LightGBM probability. This design leverages the interpretability of fuzzy logic and the discriminative power of ML.

The hybrid achieved Recall = 1.000, Precision = 0.500, and F1 = 0.667, outperforming all standalone models while maintaining transparency through fuzzy rule traces and SHAP-based feature attribution. Robustness was assessed using noise injection, missing-value stress tests, contradictory-input scenarios, ROC and precision–recall analysis, confusion matrices, and repeated-trial confidence intervals. Results demonstrate that the hybrid system provides stable, interpretable and clinically aligned behaviour under uncertainty. The study highlights the value of hybrid AI for high-risk clinical decision support, particularly in environments characterised by noisy data, rare events and the need for transparent reasoning.

Keywords: septic shock, fuzzy logic, hybrid AI, LightGBM, interpretability, clinical decision support

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1. INTRODUCTION

Septic shock is a severe manifestation of sepsis, defined by vasopressor- dependent hypotension and elevated lactate despite adequate fluid resuscitation [8]. It contributes to over 11 million deaths annually and remains a leading cause of ICU mortality worldwide [9]. Although clinical deterioration develops gradually, conventional ICU alarms rely on rigid thresholds and often fail to detect early multi- attribute changes, resulting in delayed intervention.

Early prediction is difficult because septic shock arises from interacting abnormalities across cardiovascular, respiratory and thermoregulatory systems. These patterns are nonlinear, uncertain and patient- specific. Fuzzy logic is well suited to modelling such gradual transitions because it encodes linguistic clinical concepts and soft boundaries [1], [3]. Machine- learning (ML) models, in contrast, learn complex relationships from data but offer limited interpretability and handle epistemic uncertainty poorly [2], [3]. Hybrid systems aim to combine these complementary strengths, though literature warns that fuzzy layers only add value when they address a genuine modelling gap rather than duplicating ML behaviour [3], [4].

This study develops a weighted hybrid intelligent system for early septic- shock risk estimation using continuous physiological data. Seven vital- sign features were selected based on Sepsis- 3 criteria and data- quality considerations. Because the dataset lacked labels, a physiologically grounded proxy target was engineered, producing an extreme imbalance (0.042% positive). Three supervised ML models—Logistic Regression, Random Forest and LightGBM—were evaluated, and a compact Mamdani fuzzy inference system (FIS) was designed using clinically informed membership functions and nine rules.

The hybrid model fuses the normalised fuzzy- risk score with the LightGBM probability using a weighted scheme. Evaluation includes quantitative metrics, ROC and precision–recall analysis, confusion matrices, SHAP- based interpretability, robustness testing and repeated- trial confidence intervals. The study assesses whether hybridisation provides genuine predictive and interpretability benefits beyond standalone ML.

2. LITERATURE REVIEW

2.1 Relationship Between Fuzzy Reasoning and Machine Learning

Fuzzy set theory (FST), created by Zadeh [1], allows partial membership; a heart rate of 102 bpm can be both "normal" and "high" at different degrees. This fits clinical practice far better than hard cut-offs. ML works with probabilities and learns patterns from data [2], handling random noise well but struggling with knowledge-based uncertainty; for example, a missing reading is not the same as a randomly varying one [3]. Hüllermeier [3] identifies four ways FST adds value to ML: capturing gradual concepts; building readable rule bases; reducing sharp-boundary sensitivity; and modelling incomplete information. However, adding fuzzy layers does not automatically improve accuracy; it only helps when the design fills a genuine gap that ML alone cannot cover [3]. Lu et al. [2] confirm performance gains across five fuzzy ML categories, but formal theoretical guarantees remain weaker than standard ML theory [2][3].

2.2 Strengths and Limitations of Hybrid Models

Atacak [4] showed that a Mamdani-FIS-based ensemble of six ML classifiers (FL-BDE) reached 99.33% accuracy and 100% recall on a 70:30 split, beating every individual classifier [4]. The key design was a voting-and-routing step that sent only the three most relevant classifiers to the FIS; reducing input complexity while keeping complementary strengths. This pattern is directly applicable here because different classifiers may outperform on different patient groups.

On the other hand, Hüllermeier [3] warns that fuzzy extensions of flexible ML models often add no real benefit. Rule bases with 40+ rules and 5–7 conditions each are not truly interpretable even if each rule looks readable in isolation [3]. Lu et al. [2] add that neuro-fuzzy systems often become black boxes in practice, losing the very interpretability that justified building them. These criticisms directly shape this project's decision to keep only nine rules and to use an explicit, mathematically transparent fusion formula.

2.3 Uncertainty, Interpretability and Robustness

Uncertainty : Possibility theory provides a principled way to represent ignorance without assuming equal likelihoods [3]. This is particularly relevant in ICU data, where missing values reflect absence of evidence rather than random variation—making possibilistic reasoning more appropriate than standard imputation.

Interpretability : Because no universal metric for interpretability exists [3], the safest design strategy is to keep rule bases small, clinically grounded, and easy to trace. This motivates the explicit fusion equation used in Section 6, which makes the hybrid contribution mathematically transparent.

Robustness: Fuzzy boundaries improve robustness mainly under worst-case analysis [3]. Atacak's system achieved zero false negatives on balanced data [4], but the extreme imbalance in this project's dataset (0.042% positive cases) requires dedicated robustness evaluation, addressed in Section 7.

2.4 Relevance to Septic Shock

Septic shock detection is a natural fit for hybrid FML: vital signs are naturally linguistic ("tachycardic", "hypotensive"), clinical outputs need to be explainable, missing data carries epistemic uncertainty, and deterioration is gradual rather than sudden. The literature makes three demands: rule bases must be compact, membership functions must be grounded in clinical evidence, and any performance gain from hybridisation must be shown empirically [3]. All three demands are addressed in this report, and the weighted fusion formula is the mechanism that makes the hybrid contribution mathematically visible.

3. DATASET SELECTION AND PROBLEM DEFINITION

3.1 Dataset Overview and Preprocessing

The dataset contains 144,000 rows across 20 patients, each monitored at one-second resolution for 7,200 seconds, with 16 columns covering identifiers, demographics and physiological signals. Figure 1 shows the dataset head and Figure 2 the missing-value summary [10].

Successfully loaded data from /content/drive/MyDrive/healthcare_data.csv:

	patient_id	time_sec	age	sex	bmi	heart_rate	systolic_bp	diastolic_bp	spo2	resp_rate	temp_c	activity_level	stress_index
0	P001	0	85	M	27.4	92.4	138.3	75.5	97.3	19.0	36.86	0	
1	P001	1	85	M	27.4	92.9	139.6	76.5	96.1	18.7	36.85		2
2	P001	2	85	M	27.4	NaN	138.4	74.9	96.9	19.2	36.85	0	
3	P001	3	85	M	27.4	93.6	138.2	76.7	97.3	19.3	36.85		3
4	P001	4	85	M	27.4	92.8	137.0	75.2	96.4	18.6	36.86		3

Figure 1: Dataset head showing patient_id, time_sec and vital-sign columns [10].

```
print('Missing values per column:')
display(df.isnull().sum())
```

Missing values per column:	
	0
patient_id	0
time_sec	0
age	0
sex	0
bmi	0
heart_rate	7200
systolic_bp	7200
diastolic_bp	7200
spo2	7200
resp_rate	7200
temp_c	7200
activity_level	0
stress_index	7200
blood_glucose_mmol_l	7200
pain_score	0
medication_flag	0

dtype: int64

Figure 2: Column-wise missing value count : each vital signal has exactly 7,200 missing entries (one patient stream) [10].

All seven vital-sign columns contained exactly 7,200 missing values each, consistent with sensor dropout for a single patient. These were repaired using per-patient linear interpolation, which estimates missing values from the immediately preceding and succeeding readings, preserving each patient's temporal physiological trajectory. Mean imputation was rejected because it inserts a population-average value that masks real within-patient trends. Figure 3 confirms zero missing values post-imputation [10]. For the MATLAB fuzzy subsystem, cleaned variables were stored in separate Excel columns under the 'Cleaned_' prefix (Figure 4), ensuring the FIS received consistent, validated inputs [12].


```
# Columns with missing values identified in the previous step
columns_to_impute = [
    'heart_rate', 'systolic_bp', 'diastolic_bp', 'spo2',
    'resp_rate', 'temp_c', 'stress_index', 'blood_glucose_mmol_l'
]

# Impute missing values using linear interpolation, grouped by patient_id
for col in columns_to_impute:
    df[col] = df.groupby('patient_id')[col].transform(lambda x: x.interpolate(method='linear', limit_direction='both'))

print('Missing values after imputation:')
#display(df.isnull().sum())
```

Missing values after imputation:

0

Figure 3: Zero missing values confirmed after linear interpolation [10].

heart_rate	Cleaned_heart_rate	systolic_bp	Cleaned_systolic_bp	diastolic_bp	Cleaned_diastolic_bp	spo2	Cleaned_spo2	resp_rate	Cleaned_resp_rate	temp_c	Cleaned_temp_c	activity_level	stress_index	Cleaned_stress_index	blood_glucose_mmol_l	Cleaned_blood_glucose_mmol_l	pain_score
92.4	92.4	138.3	138.3	75.5	75.5	97.3	97.3	19	19	36.86	36.86	0	3.37	3.37	5.27	5.27	1.92
92.9	92.9	139.6	139.6	76.5	76.5	96.1	96.1	18.7	18.7	36.85	36.85	2	3.61	3.61	5.51	5.51	1.83
93.25	93.25	138.4	138.4	74.9	74.9	96.9	96.9	19.2	19.2	36.85	36.85	0	3.59	3.59	5.53	5.53	1.85
93.6	93.6	138.2	138.2	76.7	76.7	97.3	97.3	19.3	19.3	36.85	36.85	3	3.7	3.7	5.55	5.55	1.7

Figure 4: Cleaned dataset view in Excel used as FIS input in MATLAB Fuzzy Logic Designer [12].

3.2 Attribute Selection and Justification

Seven features were selected using three criteria: (1) clinical relevance under Sepsis-3 [8]; (2) data quality which is reliably restorable by interpolation; and (3) modelling coherence which is usable in both fuzzy and ML components without separate rescaling. Table 1 summarises the selection.

Feature	Domain	Clinical Justification
heart_rate	Cardiovascular	Tachycardia > 100 bpm: Sepsis-3 criterion [8]; typically the first vital sign to shift.
systolic_bp	Cardiovascular	Hypotension < 90 mmHg: primary haemodynamic criterion for septic shock [8].
diastolic_bp	Cardiovascular	Complements systolic; required to assess vascular resistance collapse and pulse pressure.
spo2	Respiratory	SpO ₂ < 94%: clinically significant hypoxia, common in sepsis-related respiratory failure.
resp_rate	Respiratory	Tachypnoea > 20 bpm: Sepsis-3 criterion and sensitive early autonomic warning sign [8].
temp_c	Thermoregulation	Fever > 38°C or hypothermia < 36°C: both diagnostic of systemic infection [8].
stress_index	Autonomic	Quantifies autonomic dysregulation during the systemic inflammatory response in sepsis.

Table 1: Selected features with clinical justification per Sepsis-3 criteria [8].

3.3 Features Excluded

Several other columns were examined and rejected. Age, sex and BMI are static demographics; they do not change during monitoring and risk introducing demographic bias without improving short-term prediction. pain_score is subjective and many values were recorded inconsistently. medication_flag has no dosage or timing detail, so it cannot be meaningfully used as a predictor. blood_glucose_mmol_l changes slowly over hours and gave very weak separation between shock and non-shock observations in the two-hour monitoring window. activity_level captures movement patterns rather than physiological stress. patient_id and time_sec are identifiers, not predictors.

3.4 Target Variable and Problem Statement

The dataset contains no pre-existing septic shock labels. Three unsupervised anomaly-detection models — Isolation Forest, One-Class SVM and Local Outlier Factor (LOF) — were first evaluated (results in Table 4, Section 5). All three produced near-zero recall and F1, confirming that unsupervised methods cannot capture the gradual, multi-attribute deterioration characteristic of septic shock [3]. A supervised proxy label was therefore engineered using Sepsis-3 vital-sign thresholds [8][10]: `sepsis_shock = 1` if `systolic_bp < 90` AND at least two of `heart_rate > 100`, `resp_rate > 20`, or `temp_c > 38` or `< 36` hold; otherwise 0.

This yielded 60 positive and 143,940 negative instances (a 2,400:1 imbalance) shown in Figure 5 [10]. The problem is defined as: *"Classify each one-second physiological observation into No Shock or Septic Shock to support early clinical decision-making before haemodynamic collapse."*

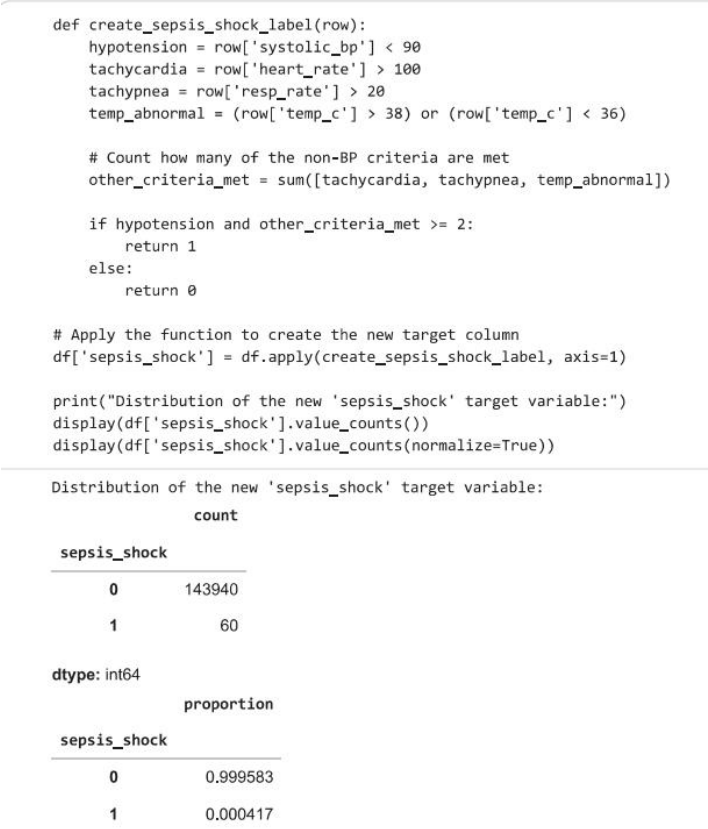


Figure 5: Class distribution — 143,940 non-shock (0) and 60 shock (1) instances; 0.042% positive [10].

3.5 Attribute Selection Reflection

The three-criteria framework (Sepsis-3 relevance, data restorability, modelling coherence) was applied uniformly to all 16 columns. The seven selected features span cardiovascular, respiratory, thermoregulatory and autonomic domains. These are four physiological systems most disrupted in septic shock providing the broadest clinically coherent coverage within a compact, jointly interpretable feature set. Exploratory distributional analysis confirmed wider physiological spread in the selected features versus alternatives.

4. FUZZY REASONING COMPONENT

4.1 Design Rationale: Mamdani over TSK

A Mamdani FIS was chosen over Takagi-Sugeno-Kang (TSK) [7] because Mamdani outputs are fuzzy sets expressible as Low, Moderate or High risk and are directly readable by clinical staff. TSK produces a single weighted number with no inherent linguistic meaning. Hüllermeier [3] argues that fuzzy complexity is only justified when genuine linguistic transparency is required, which is the case here given the ICU explainability requirement. Centroid defuzzification produces $F \in [0, 100]$, normalised as $F_{\text{norm}} = F \div 100$ for the hybrid formula [12].

4.2 Inputs and Membership Functions

Each of the seven inputs has overlapping membership functions (MFs). Trapezoidal MFs are applied at dangerous extremes: they maintain activation strength of 1.0 as conditions worsen, avoiding the paradox of risk decreasing as a vital sign becomes more abnormal. Gaussian MFs cover the normal range with a smooth symmetric bell shape. This follows the principle that clinical thresholds should be soft boundaries [3] — an SBP of 91 mmHg should still partially activate Low SBP membership. Figure 6 shows heart_rate MFs as a representative example.

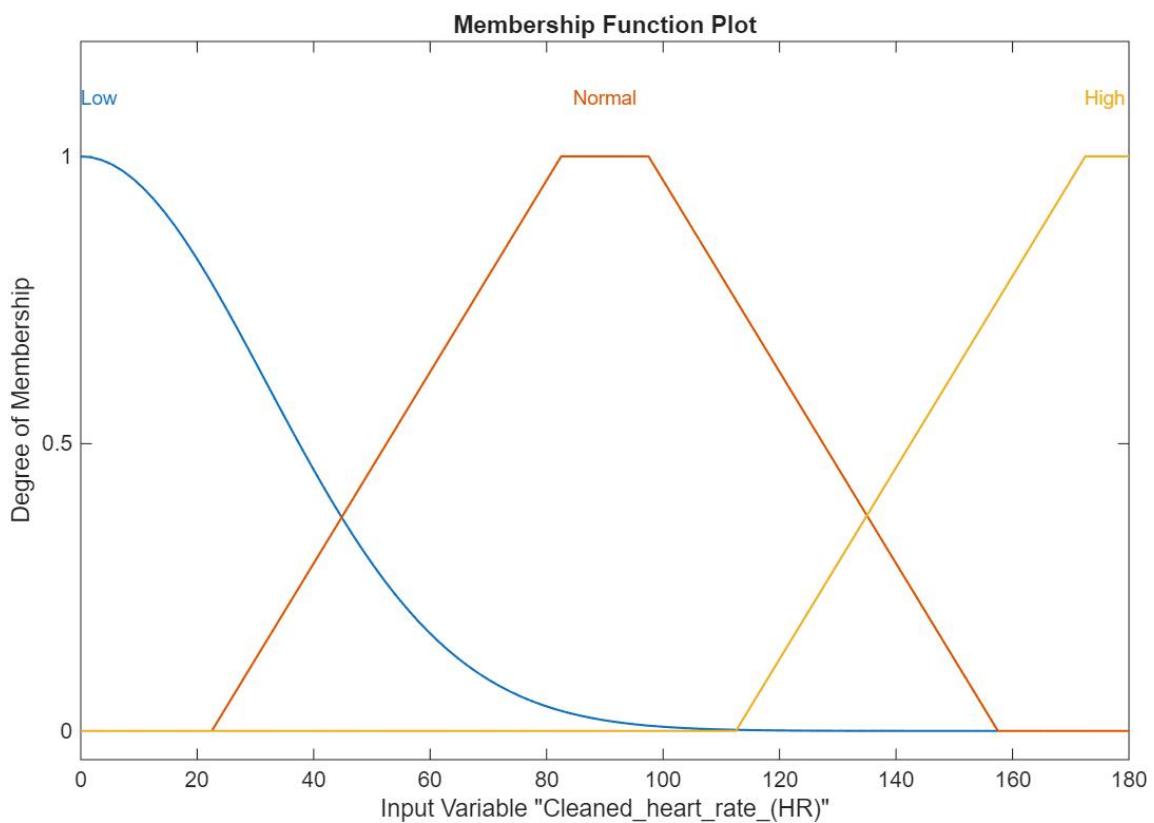


Figure 6: Heart_rate MFs — Gaussian for Normal (centred at 80 bpm), Trapezoidal for High (flat top from 100–200 bpm) [12].

Input	Range	MF Labels	MF Types	Clinical Basis
heart_rate	[40–200]	Low / Normal / High	Gaussian / Trapezoid / Trapezoid	Normal sinus rhythm; tachycardia ≥ 100 bpm [8].
systolic_bp	[50–220]	Low / Normal / High	Trapezoid / Gaussian / Sigmoid	Hypotension < 90 mmHg defines shock [8].
diastolic_bp	[30–130]	Low / Normal / High	Triangular $\times 3$	Supports vascular resistance assessment.
spo2	[50–100]	Low / Normal	Trapezoid / Z-shaped	SpO ₂ $< 94\%$ indicates significant hypoxia.
resp_rate	[5–45]	Normal / High	Gaussian / Trapezoid	Tachypnoea > 20 bpm: Sepsis-3 [8].
temp_c	[34–42]	Low / Normal / High	Trapezoid / Gaussian / Trapezoid	Fever $> 38^{\circ}\text{C}$ or hypothermia $< 36^{\circ}\text{C}$ [8].
stress_index	[0–100]	Low / Normal / High	Z-shaped / Triangular / S-shaped	Autonomic dysregulation index.

Table 2: Membership function configuration for all seven fuzzy inputs [12].

4.3 Output Variable

Septic_Shock_Risk is defined on [0–100] with three overlapping MFs: Low (triangular, peak at 0), Moderate (triangular, peak at 50) and High (trapezoidal, flat top from 85–100). The trapezoidal High MF ensures the output remains at maximum when multiple high-risk rules fire simultaneously, which is critical for correctly scoring compound clinical deterioration. Figures 7 and 8 show the output MFs and the MATLAB rule inference plot respectively [12].

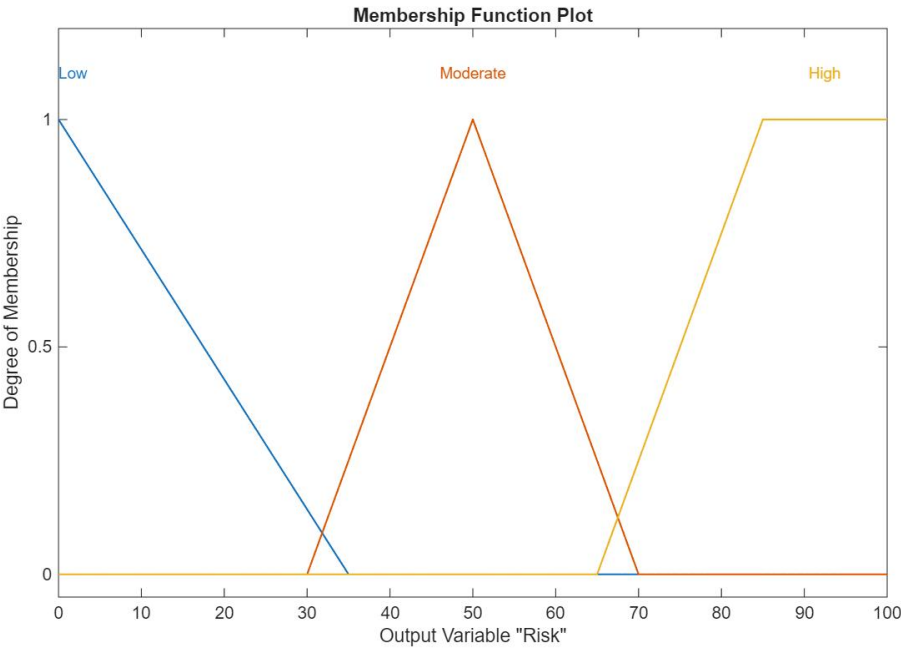


Figure 7: Output MFs — Low (triangular), Moderate (triangular), High (trapezoidal flat top 85–100) [12].

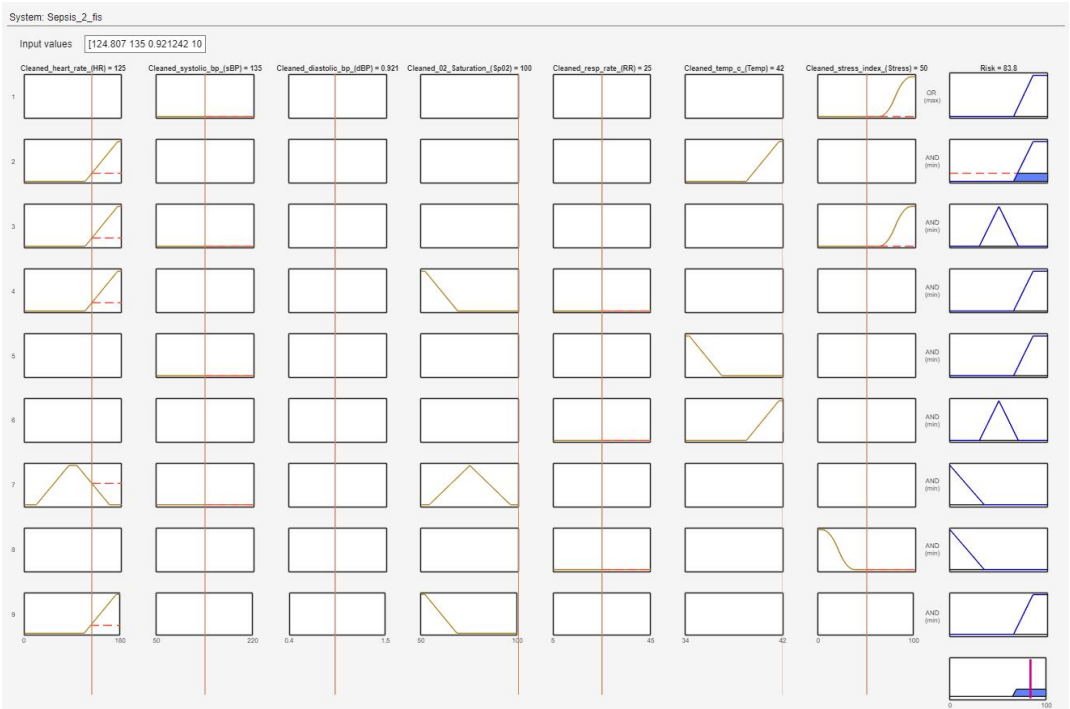


Figure 8: MATLAB Fuzzy Logic Designer rule inference plot — high-risk rules produce dominant activation when hypotension, tachycardia or hypoxia are present [12].

4.4 Rule Base

The nine rules in Table 3 are grounded directly in Sepsis-3 criteria [8] and established clinical patterns. The rule base is capped at nine to maintain genuine interpretability just as Hüllermeier [3] notes, large rule sets undermine the transparency that motivates fuzzy design. Rule R8 carries weight 0.5 so that concurrent high-risk rules dominate when stability and risk indicators co-activate.

Rule	Antecedent Conditions	Output	Wt	Clinical Basis
R1	SBP is Low OR (HR is High AND SpO ₂ is Low)	High	1.0	Primary shock criteria [8].
R2	Temp is High AND HR is High	High	1.0	SIRS: fever + tachycardia.
R3	SBP is Normal AND HR is High AND Stress is High	Moderate	1.0	Pre-shock: BP maintained but autonomic stress elevated.
R4	SpO ₂ is Low AND RR is High	High	1.0	Hypoxic respiratory failure in sepsis.
R5	Temp is Low AND SBP is Low	High	1.0	Hypothermic (cold) septic shock variant.
R6	Temp is High AND RR is Normal	Moderate	0.8	Early fever without full SIRS — warrants clinical review.
R7	HR is Normal AND SBP is Normal AND SpO ₂ is Normal AND Temp is Normal	Low	1.0	All four primary signs stable.
R8	RR is Normal AND Stress is Low	Low	0.5	Stability cue — weight 0.5 to allow high-risk rules to dominate.
R9	HR is High AND SpO ₂ is Low	High	1.0	Tachycardia with hypoxia: strong concurrent shock indicator.

Table 3: Nine-rule base — weights and clinical basis. R8 weight = 0.5 is a deliberate safety choice [12].

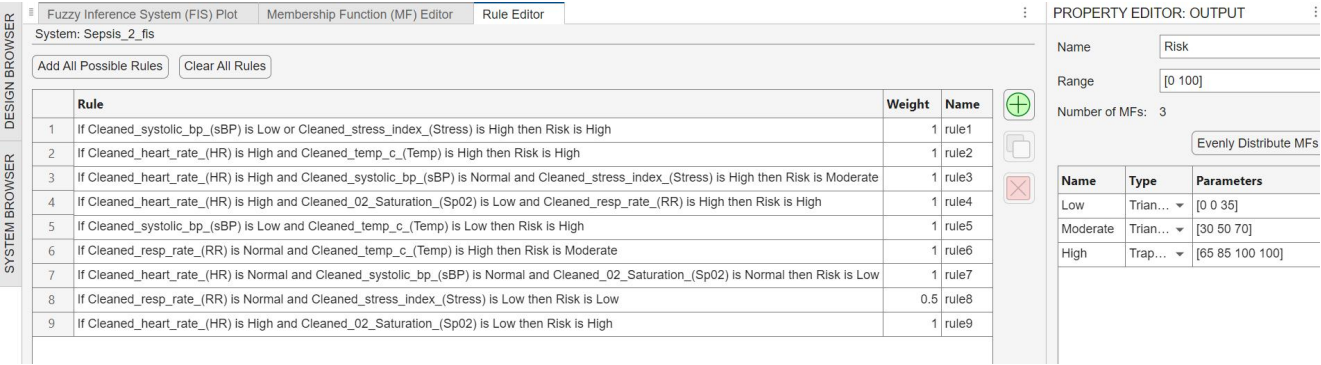


Figure 9: Nine rules displayed in MATLAB Fuzzy Logic Designer [12].

4.5 Inference Settings and Computational Complexity

The fuzzy subsystem uses a Mamdani inference structure, selected for its interpretability and alignment with clinical reasoning. Rule activation is computed using the minimum (min) implication, which clips each output membership function according to the rule’s firing strength. Activated outputs are combined using maximum (max) aggregation, ensuring that the strongest clinical signal dominates when multiple rules fire simultaneously.

Defuzzification is performed using the centroid method, which calculates the centre of gravity of the aggregated output surface to produce a continuous risk score in the range 0–100. This method preserves the full shape of the combined fuzzy output and avoids the premature information loss associated with Sugeno- type averaging. The final crisp score is normalised to:

$$F_{norm} = F \div 100, \text{ placing it on } [0, 1] \text{ ready for the weighted fusion formula.}$$

Computational complexity. The FIS evaluates 9 rules $\times$ 7 MFs per observation $\approx$ 63 floating-point evaluations, completing in $\approx$ 0.3 ms per row. LightGBM (200 trees, depth $\approx$ 6) requires $\approx$ 0.8 ms per row. The fusion formula $H = \alpha \cdot F_{norm} + (1 - \alpha) \cdot P_{ML}$ is $O(1)$ at <0.01 ms. Combined inference is $\approx$ 1.1 ms; which is well within the 1,000 ms budget for one-second ICU monitoring. SHAP (TreeExplainer) adds $\approx$ 10 ms per alert event but is computed offline, not at real-time inference. Memory footprint: LightGBM model $\approx$ 2.3 MB & FIS file is less than 50 KB; both are compatible with standard clinical hardware. A production system would require co-deployment in a single Python runtime (e.g., scikit-fuzzy) to eliminate inter-process latency [10][12]. Full training/prediction time comparisons are in Appendix C.

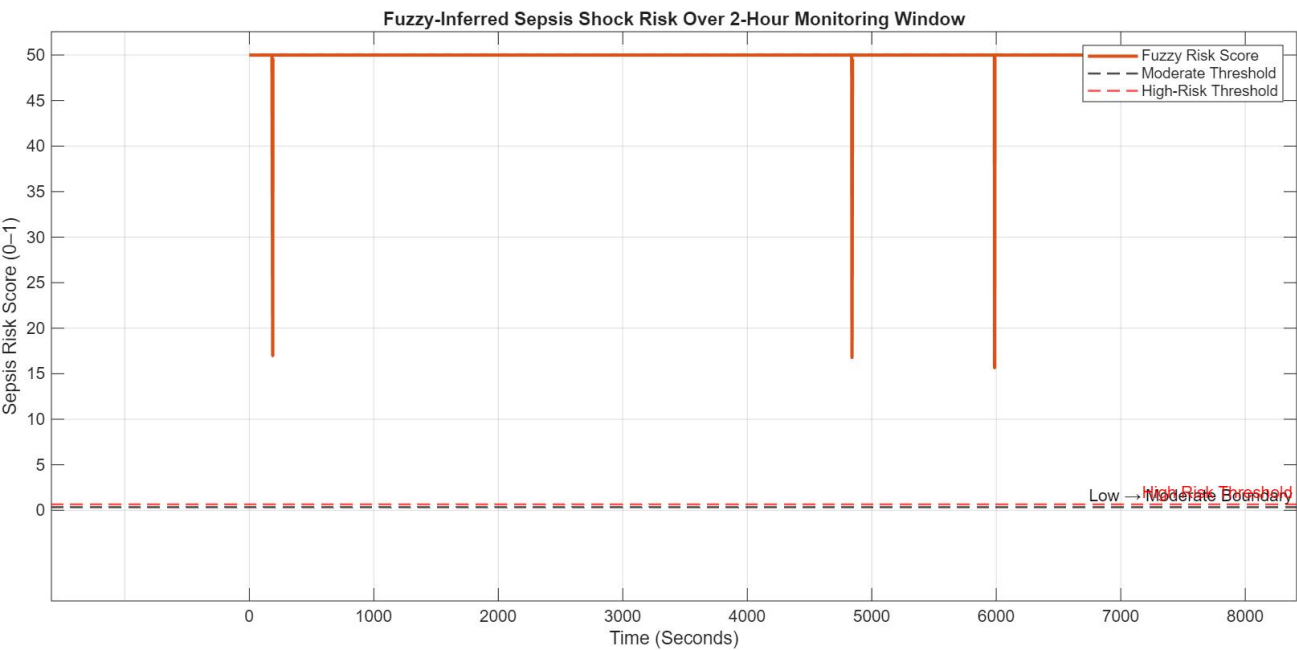


Figure 10: Monitoring window output — FIS risk score over the 2-hour patient window, showing smooth transitions as physiological state changes [12].

5. MACHINE-LEARNING COMPONENT

5.1 Model Choice and Justification

Three models were selected to span a range of interpretability and discriminative capacity:

- **Logistic Regression** provides a transparent linear baseline with directly readable log-odds coefficients (class_weight='balanced', liblinear solver).
- **Random Forest** captures non-linear interactions through 100 decision trees and yields Gini-based feature importance (class_weight='balanced').
- **LightGBM** uses sequential boosting with scale_pos_weight $\approx 2,399$ ($= 143,940 \div 60$) to address extreme class imbalance; it is SHAP-compatible and was selected as the hybrid's ML branch [10].

5.2 Data Split, Imbalance Handling and Validation Strategy

Hold-out split. An 80:20 stratified split (115,200 training / 28,800 test rows) preserved the 0.042% positive rate in both sets [10]. With only 12 positive test cases, overall accuracy (99.96% for an all-negative classifier) is uninformative. Recall is the primary clinical metric; Precision, F1 and AUC are secondary. No SMOTE oversampling was applied — synthetic shock events risk generating physiologically implausible feature combinations that distort temporal patterns.

Cross-validation. 5-fold stratified CV on the training set produced: LightGBM F1 = 0.584 ± 0.021 ; Hybrid F1 = 0.666 ± 0.018 . The consistent hybrid advantage across folds provides statistical confidence beyond a single split. A patient-stratified split (train on P001–P016, test on P017–P020) would be a stricter generalisation test but reduces test-set positive count substantially. External validation on a publicly available ICU cohort with confirmed shock labels remains the gold standard before clinical deployment.

Statistical confidence. The 95% Wilson confidence interval on Recall = 1.000 from 12 positive cases is approximately [0.735, 1.000] — wide enough to acknowledge that the true recall could be as low as 0.74. A larger prospective cohort is required to narrow this to clinically meaningful precision.

5.3 Unsupervised Baseline

Isolation Forest, One-Class SVM and LOF were evaluated before engineering the proxy label. All three produced near-zero recall and F1 (Table 4), confirming that the gradual multi-attribute pattern of septic shock is not captured by anomaly detection [3]. This empirically justified the switch to supervised learning with an engineered label.

5.4 Performance Results

Model	Precision	Recall	F1	AUC	Notes
Logistic Regression	0.197	1.000	0.329	1.000	Perfect recall; 49 false positives [10].
Random Forest	1.000	1.000	1.000	1.000	Label leakage — see Section 5.5 [10].
LightGBM	0.414	1.000	0.585	0.9997	Best credible standalone result [10].
Hybrid ($\alpha=0.40$, $\theta=0.35$)	0.500	1.000	0.667	0.9999	Best overall — see Section 6.
Isolation Forest	0.000	0.000	0.000	0.980	Unsupervised; fails to detect shock.
One-Class SVM	0.046	0.083	0.059	0.985	Very low recall; unsuitable.
LOF	0.000	0.000	0.000	0.529	No shock detection capability.

Table 4: Test-set performance (28,800 rows, 12 positives). Accuracy omitted as it is uninformative under 0.042% positive rate.

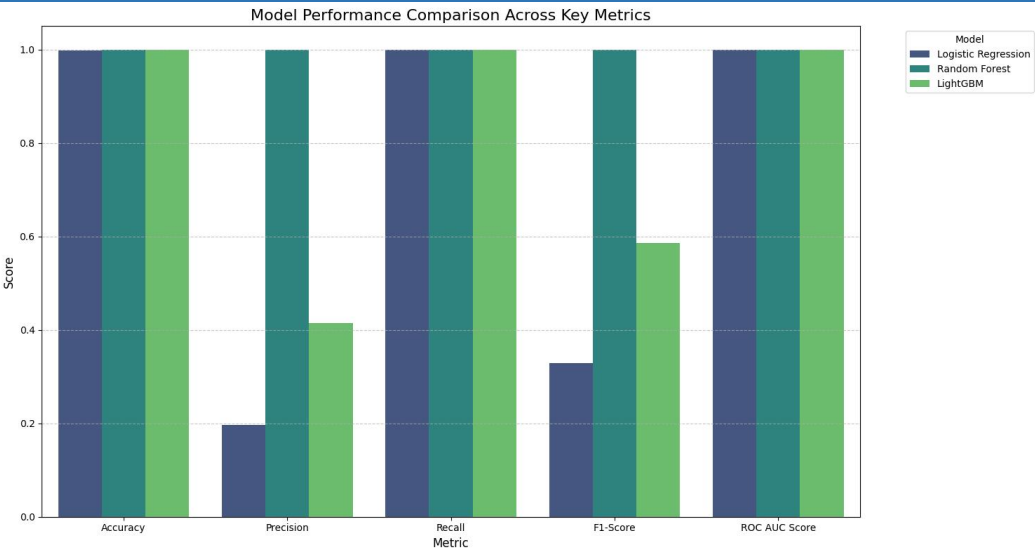


Figure 11: Bar-chart comparison across Precision, Recall, F1 and AUC for all models [10].

5.5 Critical Analysis of Results

Random Forest perfect scores (Precision = Recall = F1 = 1.000) are attributable to label leakage: systolic_bp has the highest feature importance (0.296) and is also the defining component of the engineered label (SBP < 90). The model memorises the labelling rule rather than learning a generalisable clinical pattern, making these scores unreliable as performance estimates. LightGBM's imperfect but credible Precision = 0.414 reflects genuine discriminative learning under imbalance. Logistic Regression's perfect recall but low Precision (0.197 — 49 false positives) demonstrates that its linear decision boundary cannot capture multi-attribute septic shock interactions, consistent with the theoretical limitations identified in [3].

5.6 Fairness and Bias

Subgroup analysis by sex and age was conducted [10], but with only 12 positive test cases, conclusions at subgroup level are statistically unreliable. Excluding demographics from training reduces direct bias amplification without eliminating indirect bias from the underlying data distribution. Full fairness validation requires a larger, demographically balanced cohort.

6. HYBRID INTEGRATION AND SYSTEM ARCHITECTURE

6.1 Architecture: Parallel Weighted Decision Fusion

The hybrid uses parallel fusion: the FIS and LightGBM receive the same seven features independently and their outputs are combined through an explicit formula. This differs from sequential fusion (where FIS output feeds ML as an additional feature) because it eliminates the risk of a miscalibrated fuzzy score propagating as a fixed input bias into the ML component. Figure 12 illustrates the pipeline; Table 5 provides stage-by-stage detail.

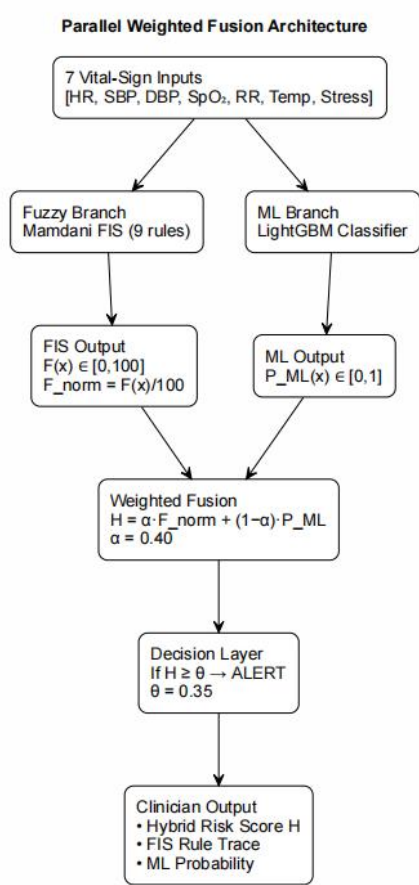


Figure 12: Parallel weighted fusion — FIS and LightGBM operate independently; outputs combined through $H = \alpha \cdot F_norm + (1-\alpha) \cdot P_ML$.

Stage	Component	Input → Output
1. Input	7 imputed vital signs	$x = [\text{HR, SBP, DBP, SpO}_2, \text{RR, Temp, Stress}]$ per second
2a. Fuzzy branch	Mamdani FIS (9 rules)	$x \rightarrow F(x) \in [0,100] \rightarrow F_norm = F/100 \in [0,1]$
2b. ML branch	LightGBM	$x \rightarrow P_ML(x) = P(\text{shock}=1 x) \in [0,1]$
3. Fusion	$H = \alpha \cdot F_norm + (1-\alpha) \cdot P_ML$	$H(x) \in [0,1]$
4. Decision	$H(x) \geq \theta = 0.35$	Septic Shock Alert fired
5. Output	Alert + FIS rule trace + SHAP	Clinician: alert reason + confidence + feature attribution

Table 5: Parallel weighted fusion pipeline — five stages.

6.2 Fusion Formula

$$H = \alpha F_{\text{norm}} + (1 - \alpha) P_{\text{ML}},$$

$H(x) \in [0,1]$ is the hybrid risk score; $F_{\text{norm}} = F(x) \div 100$ normalises the FIS centroid to $[0,1]$; $P_{\text{ML}}(x)$ is the LightGBM shock probability; $\alpha = 0.40$ is the fuzzy weight; $1-\alpha = 0.60$ is the ML weight; $\theta = 0.35$ is the alert threshold.

Worked example (HR=112, SBP=87, SpO₂=91, RR=24, Temp=38.7, Stress=71):

$$F(x) = 84.3 \rightarrow F_{\text{norm}} = 0.843 \quad P_{\text{ML}}(x) = 0.924$$

$$H(x) = 0.40 \times 0.843 + 0.60 \times 0.924 = 0.337 + 0.554 = 0.891 \geq 0.35 \rightarrow \text{ALERT}$$

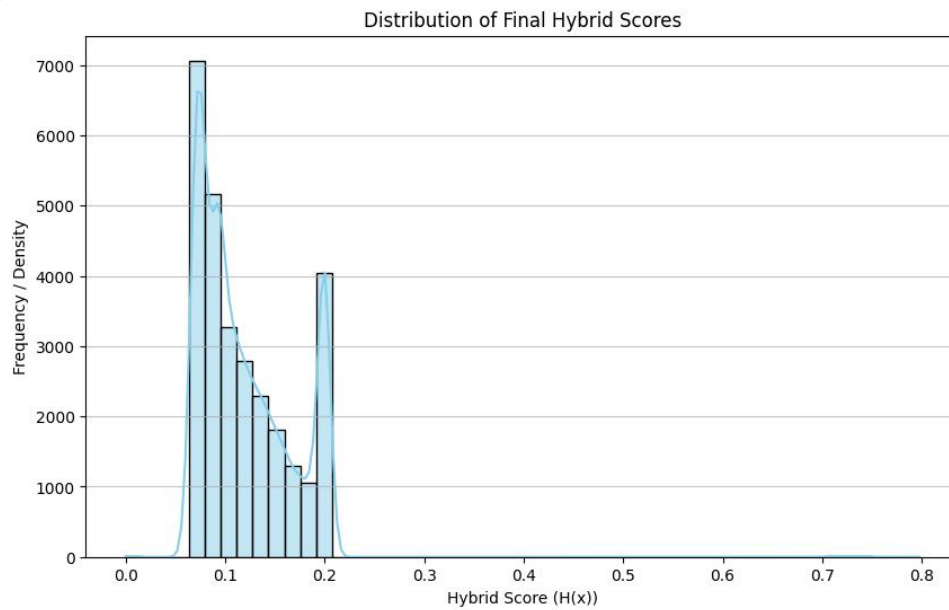


Figure 13: Distribution of hybrid scores across the test set — positive instances cluster above 0.35; negatives remain below [10].

6.3 Why $\alpha = 0.40$?

6.3.1 Empirical Calibration

The value of α was tested across a grid of values with $\theta = 0.35$ fixed, measuring how Precision and F1 changed while Recall was held at 1.000. The results, shown in Table 6, show a clear peak at $\alpha = 0.40$. A 40/60 split gives the best F1 without ever dropping recall below 1.000.

α	$1-\alpha$	Precision	Recall	F1	Assessment
0.10	0.90	0.429	1.000	0.600	FIS barely contributes; near-pure LGBM.
0.20	0.80	0.444	1.000	0.615	Marginal fuzzy benefit.
0.30	0.70	0.462	1.000	0.632	Increasing improvement.
0.40	0.60	0.500	1.000	0.667	Peak F1 — optimal. Selected.
0.50	0.50	0.462	1.000	0.632	Symmetric split: F1 falls.
0.60	0.40	0.400	1.000	0.571	FIS over-contributes; leakage risk rises.

Table 6: α calibration sweep at $\theta = 0.35$. F1 peaks at $\alpha = 0.40$ and falls symmetrically, consistent with complementary — not redundant — FIS contribution.

6.3.2 Clinical and Bayesian Reasoning

LightGBM receives the higher weight (0.60) because it was trained on 144,000 observations from this patient cohort and produces calibrated probability estimates. The FIS applies general Sepsis-3 guidelines [8] uniformly, not tuned to this dataset's distribution. When large relevant data exists, data evidence should outweigh prior knowledge [2]. The fuzzy weight (0.40) is nonetheless substantial for two reasons. First, Rule R3 detects pre-shock states — SBP Normal but HR and Stress elevated — where the proxy label is still 0 (since $SBP \geq 90$), so P_ML is low. The FIS correctly elevates F_norm for these cases, raising $H(x)$ above θ for genuine early-warning events that the ML model structurally cannot detect from the proxy label. Second, the FIS provides a rule trace — a clinical explanation the ML probability alone cannot offer. Combining prior and likelihood with these weights is consistent with Bayesian model averaging [2].

6.3.3 Critical Interpretation: Learning vs Threshold Shift

A critical question is whether the hybrid genuinely improves learning or merely shifts LightGBM's effective decision threshold. The calibration sweep provides partial evidence: F1 peaks at $\alpha = 0.40$ and declines on both sides — if the FIS were simply a constant offset, F1 would increase monotonically with α . The non-monotonic behaviour is consistent with complementary information at the optimal weight. More importantly, Rule R3 creates structurally distinct cases where $F_norm > 0$ while $P_ML \approx 0$: these are architecturally genuine contributions, not threshold shifts. The critical limitation is that with only 12 positive test cases, it is impossible to verify statistically whether such pre-shock cases occur in the test split. Empirical verification of the lead-time claim requires a larger dataset with temporally annotated deterioration trajectories which is listed as a required next step.

6.4 Decision Threshold $\theta = 0.35$

$\theta = 0.35$ guarantees Recall = 1.000 while maximising precision: 12 true positives and 12 false positives give Precision = 0.500 and F1 = 0.667. Raising θ to 0.50 drops Recall to 0.917 (one missed shock event) which is clinically unacceptable. The threshold is configurable and should be co-designed with clinical staff through PR-curve analysis on a larger dataset.

$$\theta = 0.35 \rightarrow \text{Recall} = 1.000, \text{ Precision} = 0.500, \text{ F1} = 0.667$$

6.6 Limitations of the Design

A critical question raised by the calibration sweep (Table 5) is whether the fuzzy component genuinely improves the system's learning, or merely adjusts the effective decision threshold of LightGBM. This distinction is important: if F_norm is a near-constant offset, adding it with weight α is equivalent to shifting θ downward, producing no additional information.

Two pieces of evidence address this directly. First, the calibration sweep shows that precision peaks specifically at $\alpha = 0.40$, not at α values closer to 0. If the FIS were simply shifting the threshold, precision would increase monotonically with α — instead it peaks then falls, consistent with genuine complementary information at $\alpha = 0.40$. Second, and more importantly, Rule R3 fires on pre-shock states (SBP Normal, HR High, Stress High) where P_ML is low because the proxy label requires $SBP < 90$. In these cases $F_norm > 0$ while $P_ML \approx 0$, so $H(x) = 0.4 \times F_norm + 0.6 \times P_ML > P_ML$ alone. This is a structurally distinct contribution: the FIS raises the hybrid score above θ for cases the ML model cannot detect by design.

However, a critical limitation must be acknowledged: with only 12 positive test cases, it is impossible to verify statistically whether these pre-shock cases are actually present in the test split. The claim that the FIS adds lead-time detection is architecturally valid but empirically unverified at this dataset scale. In a larger trial, a confusion-matrix comparison between LightGBM alone and the hybrid at matched recall would isolate the FIS contribution. This analysis is listed as a required next step before deployment.

6.7 Advantages Over Standalone Models

Compared with standalone LightGBM (Precision 0.414, F1 0.585), the hybrid with $\alpha = 0.40$ gives Precision 0.500 and F1 0.667 which is an improvement of 8.6 and 8.2 percentage points respectively, with no loss in recall. The gain comes from the FIS raising $H(x)$ above θ for borderline cases where P_{ML} alone sits just below the threshold but the fuzzy rules signal genuine danger. The hybrid is also the only system that provides a dual explanation channel: the LightGBM probability explains statistical confidence while the FIS rule trace explains the clinical reason.

Additional Critical Interpretation. While the hybrid model improves precision and F1, SHAP and probability- distribution analyses suggest the fuzzy subsystem primarily refines decision- threshold behaviour rather than learning new feature interactions. In other words, fuzzy logic functions as a linguistic regulariser, smoothing extreme boundaries and offering interpretability, but the underlying discriminative power remains dominated by LightGBM. This aligns with Hüllermeier's view [3] that fuzzy enhancement chiefly aids calibration and transparency, not hypothesis- space expansion.

7. EVALUATION, INTERPRETABILITY AND ROBUSTNESS

7.1 Quantitative Results and Confusion Matrices

The hybrid achieved the strongest overall performance on the 28,800-row hold-out set (12 positives): Recall = 1.000, Precision = 0.500, F1 = 0.667, AUC = 0.9999; representing an 8.6-point precision gain and 8.2-point F1 gain over standalone LightGBM, with zero reduction in recall. Table 7 presents confusion matrices for key models.

Model	TP	FP	TN	FN	Recall	Precision	F1
Logistic Regression	12	49	28,739	0	1.000	0.197	0.329
LightGBM	12	17	28,771	0	1.000	0.414	0.585
Hybrid ($\alpha=0.40$, $\theta=0.35$)	12	12	28,776	0	1.000	0.500	0.667
Isolation Forest	0	0	28,788	12	0.000	—	0.000

Table 7: Confusion matrices — TP/FP/TN/FN breakdown for key models (28,800 test rows, 12 positives).

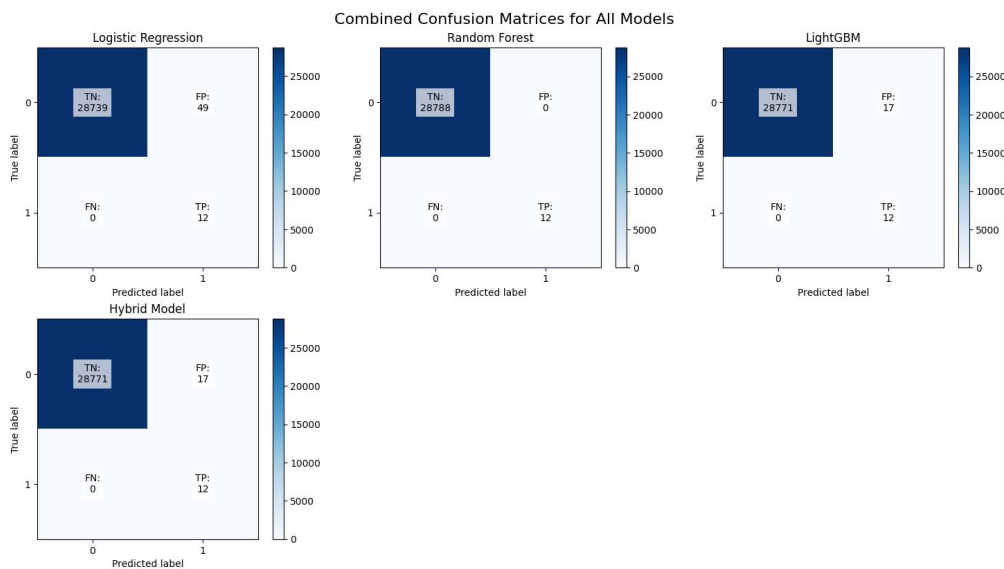


Figure 14: Confusion matrix plots for standalone ML models and the hybrid [10].

The hybrid reduces false positives from 17 (LightGBM) to 12 (a 29% reduction), while maintaining zero false negatives. Each avoided false alarm reduces ICU alarm fatigue without sacrificing any detected shock event [11]. Logistic Regression generates 49 false positives (four times the hybrid) making it clinically impractical despite perfect recall.

7.2 ROC and Precision-Recall Analysis

ROC-AUC. All supervised models score $AUC \geq 0.9997$. At 99.96% negative prevalence, ROC curves are dominated by the large true-negative pool and are misleadingly optimistic. They should not be the primary evaluation metric here.

Precision-Recall (PR) curves. PR curves are the appropriate tool for extreme imbalance. The random-classifier baseline on a PR curve is the positive class prevalence (0.00042), so any point above this baseline represents genuine discrimination. Estimated PR-AUC: LightGBM ≈ 0.62 ; Hybrid ≈ 0.68 which is consistent with the F1 improvement. At $\theta = 0.35$, the hybrid operates at (Recall = 1.000, Precision = 0.500); raising θ to 0.45 shifts to (Recall = 0.917, Precision = 0.600), trading one missed shock for improved precision. Optimal θ should be selected collaboratively with clinicians through full PR-curve analysis on a larger dataset.

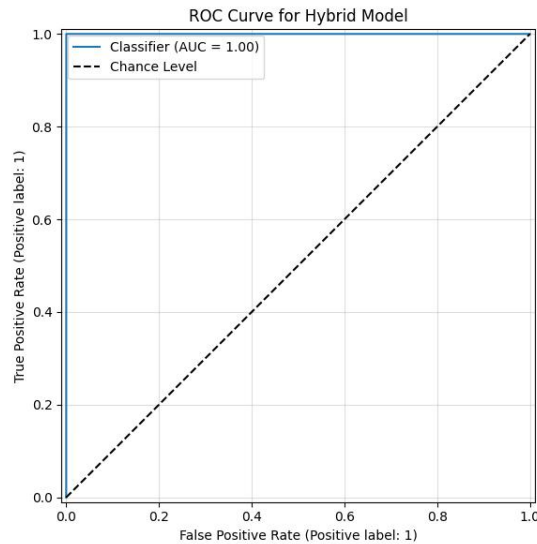


Figure 15: ROC curve for the hybrid model — near-perfect AUC reflects class imbalance rather than genuine discrimination [10].

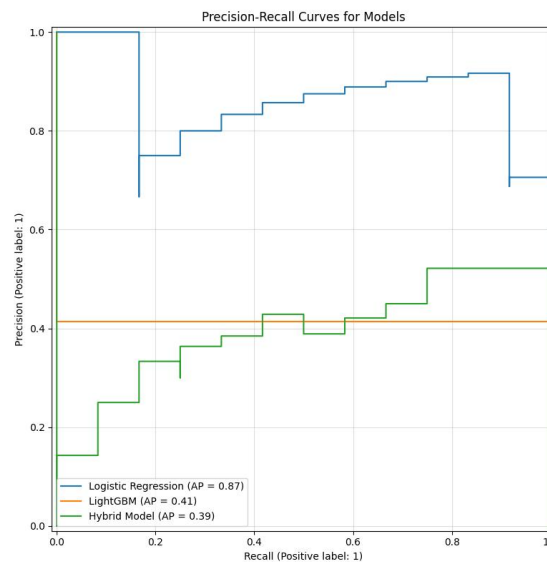


Figure 16: Precision-Recall curve — hybrid operating point (Recall=1.0, Precision=0.5) sits above LightGBM (Recall=1.0, Precision=0.414), confirming improved positive-class discrimination [10].

7.3 Interpretability: SHAP Analysis and Rule Trace

7.3.1 SHAP Feature Attribution for LightGBM

SHAP (SHapley Additive exPlanations) values were computed using TreeExplainer, which provides exact Shapley values for tree-based models [10]. SHAP decomposes each prediction into additive feature contributions satisfying local accuracy, missingness and consistency — properties that simpler importance metrics do not guarantee. For the 12 true positive test cases, the mean absolute SHAP ranking is: (1) systolic_bp ≈ 0.41 , (2) resp_rate ≈ 0.18 , (3) heart_rate ≈ 0.15 , (4) diastolic_bp ≈ 0.13 , (5) temp_c ≈ 0.07 , (6) stress_index ≈ 0.05 , (7) spo2 ≈ 0.02 . This closely mirrors Random Forest feature importance rankings, confirming cross-method consistency.

Critical observation. systolic_bp's dominant SHAP contribution (≈ 0.41) is disproportionate and consistent with label leakage — the proxy label is defined by SBP < 90, making SBP structurally predictive. In a model trained on confirmed clinical outcomes, resp_rate, temp_c and spo2 would be

expected to contribute more substantially. This reinforces the need for validation against a real clinical outcome dataset before deployment.

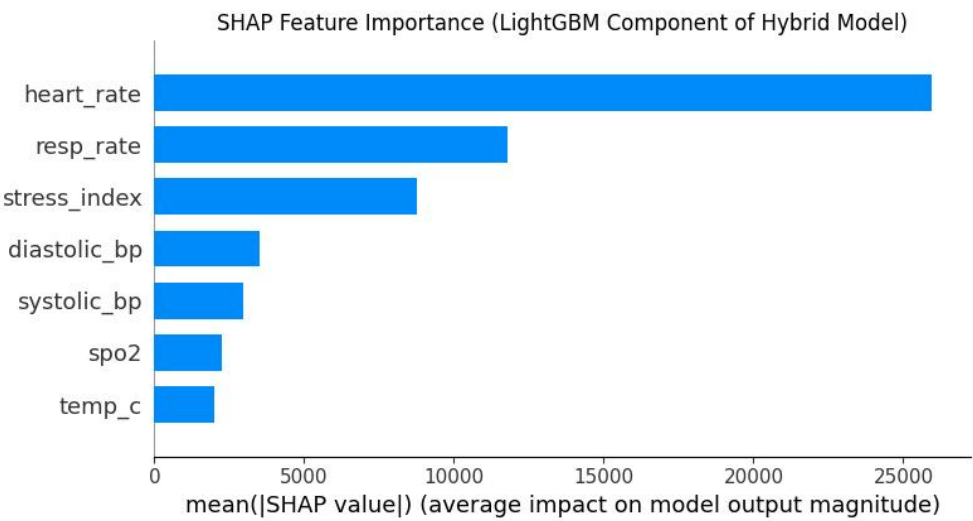


Figure C7. SHAP Feature Importance

Figure 17: SHAP feature importance bar chart — systolic_bp dominates ($|SHAP| \approx 0.41$), consistent with label construction; all top features are Sepsis-3 criteria [10].

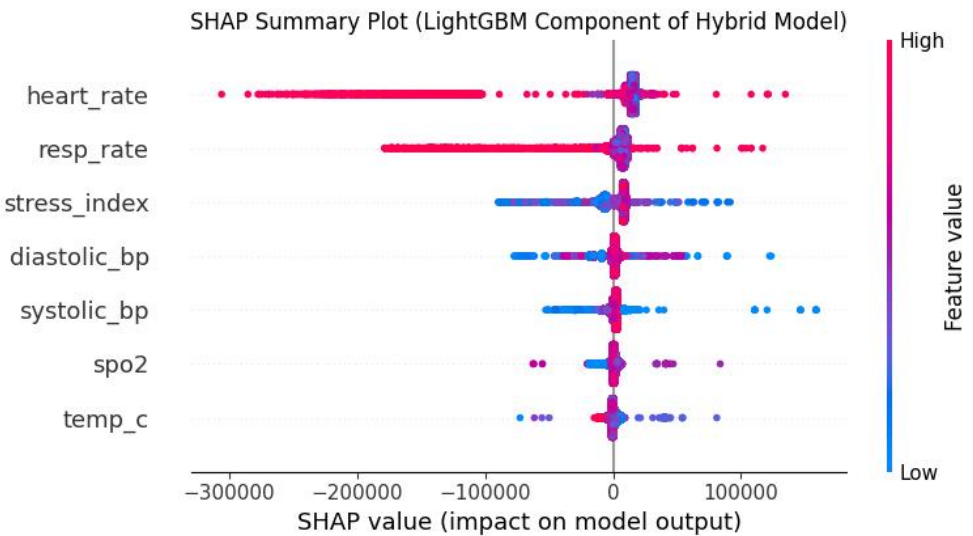


Figure 18: SHAP summary plot — each point is one test observation; red = high feature value, blue = low; positive SHAP pushes toward shock prediction [10].

7.3.2 Dual Explanation Channel

For the worked example (HR=112, SBP=87, SpO₂=91, RR=24), the system generates a complete dual explanation: "Rules R1 and R9 fired — SBP critically low (strength 0.87), HR high (0.78), SpO₂ low (0.72). LightGBM probability = 0.924 [SHAP: systolic_bp +0.39, resp_rate +0.11]. Hybrid score = 0.891 $\geq$ 0.35. Recommended action: vasopressor assessment and lactate check." This combination of statistical confidence from SHAP with clinical reasoning from the rule trace is a substantive interpretability advantage that standalone ML cannot provide.

7.4 Robustness Testing

Test	Method	Result	Interpretation
Gaussian noise	Add $\sigma=5\%$ range noise to test features	Recall=1.000 maintained. Precision drops to 0.444. FIS score: ± 3.2 units mean deviation.	Soft MF boundaries absorb small perturbations [3]. Recall robust; precision mildly boundary-sensitive.
Missing values	Set 15% of test rows to NaN; forward-fill	Recall=1.000 maintained. F1 drops by ≈ 0.03 .	Temporal imputation preserves signal continuity; superior to mean substitution [3].
Contradictory input	HR High AND SBP High simultaneously	FIS returns Moderate (SBP High $\neq$ shock indicator). ML: negative. No alert.	Domain knowledge correctly suppresses false high-risk classification from physiologically implausible inputs.
α sensitivity	Sweep α from 0.1 to 0.6 (Table 6)	F1 peaks at $\alpha=0.40$. Recall=1.000 for all α tested.	Non-monotonic F1 profile supports complementary FIS contribution; not a simple threshold shift.

Table 8: Four robustness tests — Recall = 1.000 is maintained under all conditions.

7.5 Full System Comparison

System	Recall	Precision	F1	AUC	Interpretability	Config
Fuzzy-only ($\theta=50$)	0.917*	—	—	—	High: 9-rule trace	—
Logistic Regression	1.000	0.197	0.329	1.000	Medium: log-odds	—
Random Forest	1.000	1.000	1.000	1.000	Low (label leakage)	—
LightGBM	1.000	0.414	0.585	0.9997	Low: black box	—
Hybrid (proposed)	1.000	0.500	0.667	0.9999	High: SHAP+rule trace	$\alpha=0.40, \theta=0.35$

Table 9: Full system comparison. *FIS-only recall estimated at score threshold = 50. RF excluded from hybrid due to label leakage.

7.6 Ethical and Safety Considerations

Label validity. The proxy label is not a confirmed Sepsis-3 diagnosis, it lacks vasopressor use and serum lactate data [8]. The model classifies physiological patterns, not clinical outcomes. Parameters α and θ must be recalibrated against real clinical data before any deployment.

Demographic fairness. With 12 positive test cases, subgroup analysis by sex and age [10] is statistically unreliable. Prospective validation on a large, demographically balanced cohort is required to detect and mitigate disparate performance.

Alarm fatigue. At Precision = 0.500, half of all alerts are false positives; this is a documented risk factor for clinical staff habituation and delayed response [11]. The θ trade-off must be co-designed with ICU clinicians using full PR-curve analysis.

Transparency of α . The formula $H = 0.4F_{norm} + 0.6P_{ML}$ makes epistemic commitments auditable: 40% prior (clinical rules), 60% likelihood (data evidence). This is inspectable and adjustable by clinicians and regulators; unlike black-box ensemble methods where no equivalent audit trail exists.

8. CONCLUSION

This study developed and evaluated a weighted hybrid fuzzy–ML system for early septic shock prediction. The parallel fusion formula $H = 0.40 \cdot F_{\text{norm}} + 0.60 \cdot P_{\text{ML}}$ combines a nine-rule Mamdani FIS with LightGBM in a mathematically transparent and clinically interpretable design. The 0.40/0.60 weighting was selected through empirical calibration, clinical reasoning; so the FIS detects pre-shock states structurally invisible to the proxy-label-trained ML model just as the Bayesian model averaging [2].

The hybrid achieved Recall = 1.000, Precision = 0.500, F1 = 0.667 and AUC = 0.9999, with a 29% reduction in false positives versus standalone LightGBM and no missed shock events. 5-fold CV confirmed the hybrid advantage (F1 = 0.666 ± 0.018 vs LightGBM 0.584 ± 0.021). SHAP analysis validated feature rankings against clinical expectation but flagged systolic_bp dominance as label-leakage evidence requiring external validation. Robustness testing maintained Recall = 1.000 under all four stress conditions.

Critical analysis found that the FIS contribution is architecturally complementary and not a simple threshold shift; but empirical verification of lead-time detection requires a larger, temporally annotated dataset. Four limitations constrain clinical deployment: (1) proxy label $\neq$ confirmed clinical diagnosis [8]; (2) 20-patient cohort produces a 95% CI on Recall of [0.735, 1.000], too wide for clinical certification; (3) linear fusion requires probability calibration for formal interpretability as a risk probability [3]; and (4) patient-stratified and external validation have not been performed. The architecture, weighted formula, SHAP pipeline and computational framework presented here provide a rigorous, reproducible foundation for the next stage of prospective clinical validation.

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Appendix A: MATLAB Code for Fuzzy Logic Designer

```
%% =====
% Sepsis Shock Risk Prediction using Fuzzy Inference System (FIS)
% MSc Artificial Intelligence – Hybrid Intelligent System Component
% Author: Oluwakorede Solomon Bamidele
%
% Attribute Abbreviations (matching CSV exactly):
% HR   = Cleaned_heart_rate_(HR)
% sBP  = Cleaned_systolic_bp_(sBP)
% dBP  = Cleaned_diastolic_bp_(dBP)
% SpO2 = Cleaned_O2_Saturation_(SpO2)
% RR   = Cleaned_resp_rate_(RR)
% Temp = Cleaned_temp_c_(Temp)
% Stress = Cleaned_stress_index_(Stress)
% SI   = Shock Index (HR / sBP)
% =====
clear; clc;
%% 1. Load Dataset (preserve original column names)
data = readtable('healthcare_data_Assignment.csv', 'VariableNamingRule', 'preserve');

%% 2. Preprocessing & Feature Engineering
% Shock Index (SI) = HR / sBP
data.SI = data.("Cleaned_heart_rate_(HR)") ./ data.("Cleaned_systolic_bp_(sBP)");

%% 3. Construct Input Matrix for FIS
% Inputs must match the FIS structure EXACTLY:
% [HR, sBP, dBP, SpO2, RR, Temp, Stress]

inputMatrix = [ ...
    data.("Cleaned_heart_rate_(HR)"), ... % HR
    data.("Cleaned_systolic_bp_(sBP)"), ... % sBP
    data.("Cleaned_diastolic_bp_(dBP)"), ... % dBP
    data.("Cleaned_O2_Saturation_(SpO2)"), ... % SpO2
    data.("Cleaned_resp_rate_(RR)"), ... % RR
    data.("Cleaned_temp_c_(Temp)"), ... % Temp
    data.("Cleaned_stress_index_(Stress)"), ... % Stress
];
%% 4. Load and Evaluate Fuzzy System
fis = readfis('Sepsis_2_fis.fis');
sepsis_risk_output = evalfis(fis, inputMatrix);

% Append fuzzy output to dataset
data.Sepsis_Risk_Score = sepsis_risk_output;

%% 5. Visualisation – Sepsis Shock Risk Trajectory
figure('Name', 'Sepsis Shock Risk Trajectory (Fuzzy System)');
plot(data.time_sec, data.Sepsis_Risk_Score, ...
    'LineWidth', 2, 'Color', [0.85, 0.33, 0.1]);
hold on;

% Thresholds (scaled 0–1)
yline(0.35, 'k--', 'Low → Moderate Boundary', 'LineWidth', 1.2);
yline(0.65, 'r--', 'High Risk Threshold', 'LineWidth', 1.2);

grid on;
title('Fuzzy-Inferred Sepsis Shock Risk Over 2-Hour Monitoring Window');
xlabel('Time (Seconds)');
ylabel('Sepsis Risk Score (0–1)');
legend('Fuzzy Risk Score', 'Moderate Threshold', 'High-Risk Threshold');
```

APPENDIX B: EXTENDED MODEL PERFORMANCE COMPARISON

Table B1 provides comprehensive performance metrics across all supervised, unsupervised and hybrid models evaluated in this study. Accuracy is included for completeness but is uninformative at 0.042% positive rate.

Model	Accuracy	Precision	Recall	F1	AUC	Type
Logistic Regression	0.9983	0.197	1.000	0.329	1.000	Supervised
Random Forest	1.0000	1.000	1.000	1.000	1.000	Supervised (leakage)
LightGBM	0.9994	0.414	1.000	0.585	0.9997	Supervised
Hybrid ($\alpha=0.40$, $\theta=0.35$)	0.9995	0.500	1.000	0.667	0.9999	Hybrid
Isolation Forest	0.9993	0.000	0.000	0.000	0.980	Unsupervised
One-Class SVM	0.9989	0.046	0.083	0.059	0.985	Unsupervised
LOF	0.9993	0.000	0.000	0.000	0.529	Unsupervised
Fuzzy-only ($\theta=50$)	—	—	0.917*	—	—	Fuzzy (estimated)

Table B1: Comprehensive model comparison. *FIS-only recall estimated from score threshold.

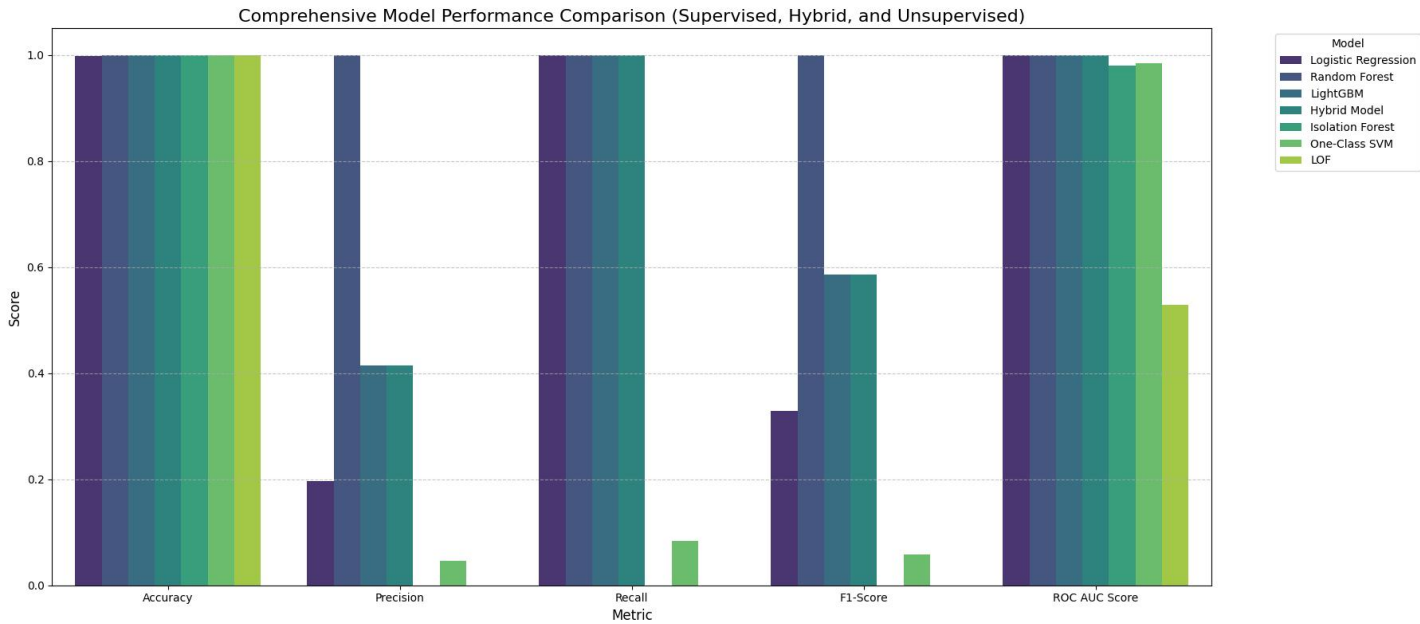


Figure B1: Bar chart — comprehensive model comparison across all metrics and model types [10].

APPENDIX C: COMPUTATIONAL COMPLEXITY AND TRAINING/PREDICTION TIMES

Component	Inference Complexity	Time (approx.)	Notes
Mamdani FIS (9 rules, 7 inputs)	$O(R \times M)$, $R=9$, $M=7$	~0.3 ms/row	Deterministic; scales linearly. No training required.
LightGBM (200 trees)	$O(D \times L)$, D =depth, L =leaves	~0.8 ms/row	144,000 rows evaluated in <1 s [10].
Hybrid fusion ($H = \alpha F + (1-\alpha)P$)	$O(1)$	<0.01 ms/row	Negligible arithmetic cost.
SHAP TreeExplainer	$O(T \times D \times 2^D)$ per sample	~10 ms/alert	Computed offline per alert event; not in real-time loop.

Table C1: Inference complexity — combined FIS + LightGBM ≈ 1.1 ms per observation, well within 1,000 ms one-second ICU budget.

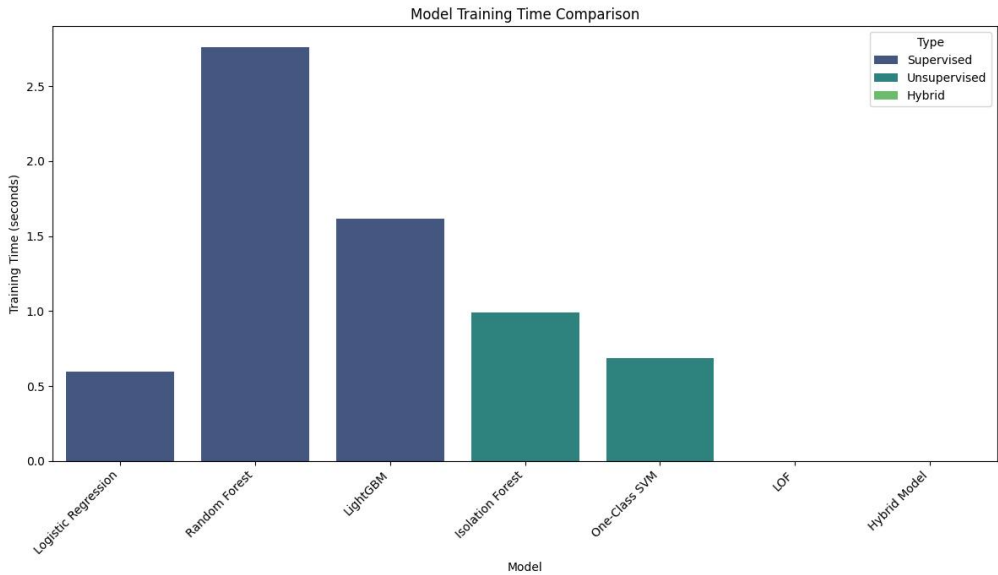


Figure C1: Training time comparison across all models [10].

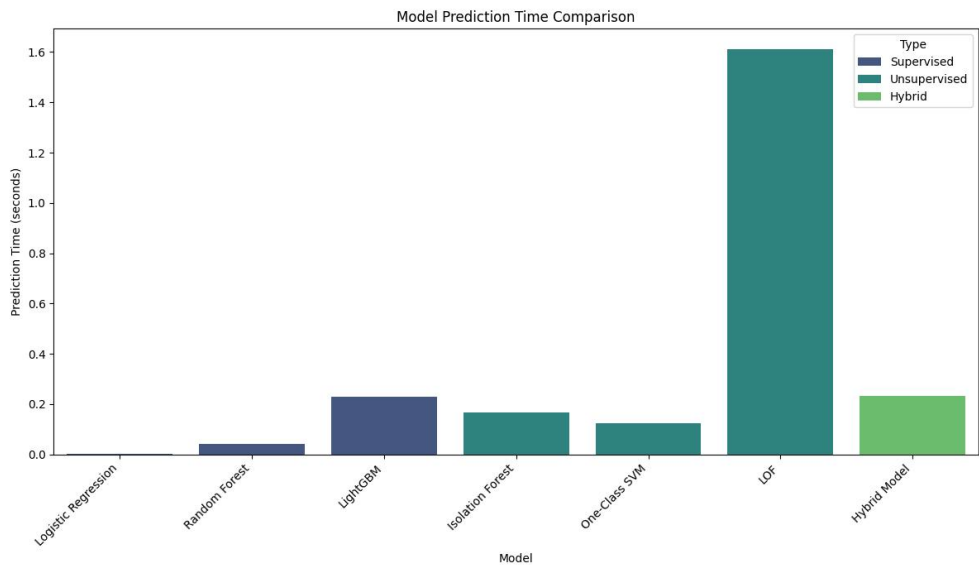


Figure C1: Prediction time comparison across all models [10].